



LABORATORY MANUAL

MES210TC

Continuous and Discrete Time Signals and Systems

Student Name:

ID Number:

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Experiment

This experiment uses MATLAB to observe and analyze fundamental concepts in Signals and Systems. You are expected to perform simulations, generate plots, and verify theoretical principles using MATLAB. Your submitted report must include relevant screenshots of your MATLAB code (.m files) and the resulting figures, along with brief descriptions of your observations and outcomes.

1. The core concepts and objectives of this experiment are outlined in the provided lab manual and lecture materials.
2. For general guidance on using MATLAB, please refer to the official MATLAB website or its comprehensive documentation.
3. To learn the usage of specific MATLAB functions or commands, use the help command within MATLAB (e.g., help fft) or explore the built-in examples.

For the experiment, every student needs to submit **one file** that includes the **program code**, **results**, and **relevant descriptions**.

Deadline: Sunday 7 June

Late submission will follow university policy.

Note: Please make sure that your **name and student ID** are written on your Lab report cover page.

Experiment 1 (40 marks)**Experiment 1.1** Generation and Visualization of Basic Continuous-Time Signals Using MATLAB (10 marks)

Objective: To master the method of representing basic continuous-time signals using MATLAB and plotting their waveforms.

Tasks:

1. Use MATLAB commands to generate the following signals and plot their waveforms. Ensure the plots have clear x-axis and y-axis labels.

(1) Single-sided exponential decay signal: $x_1(t) = 3e^{-2t}u(t)$.

$u(t)$ is unit step function.

(2) Sinusoidal signal: $x_2(t) = 1.5\sin(4\pi t - \pi/3)$

(3) Rectangular pulse signal:

$$x_3(t) = \begin{cases} 1.5, & 0 \leq t \leq 2 \\ 0, & \text{Otherwise} \end{cases}$$

After plotting, use the axis command to adjust the axis limits so that the waveform is clearly displayed in the center of the figure window.

(4) Tremolo Effect Demonstration: (10 marks)

The tremolo effect can be produced by multiplying a high-frequency sinusoidal signal (carrier) with a low-frequency sinusoidal signal (modulator). Generate the signal $x_4(t) = \sin(2\pi \cdot 880t) \cdot (1 + 0.7\sin(2\pi \cdot 5t))$, where $0 \leq t \leq 3$ seconds. Plot its waveform for the first 0.5 seconds, and use the **sound** function to play the entire 3-second audio (suggested sampling frequency $f_s \geq 8192$ Hz).

Experiment 1.2: Basic Operations on Continuous-Time Signals
(10 marks)

Objective: To master the basic operations on continuous-time signals, including time shifting, scaling, and reversal, and visualize their effects on signal waveforms using MATLAB.

Given Signal:

Consider a piecewise linear signal $f(t)$ defined as:

$$f(t) = \begin{cases} t + 1, & -1 \leq t \leq 0 \\ 1 - t, & 0 < t \leq 1 \end{cases}$$

Tasks:

1. Plot the Original Signal: Use MATLAB commands to plot the waveform of $f(t)$ over the time range from $t=-3$ to $t=3$. Add grid lines, and label the x-axis and y-axis with appropriate titles. (2 marks)

2. Basic Signal Operations: In the same figure window, use the **subplot** command to plot the following transformed signals (over the same time range from $t=-3$ to $t=3$): (6 marks)

(1) $f(t-2)$: Shifted right by 2 units

(2) $f(3t)$: Compressed by a factor of 3 on the time axis

(3) $f(-t)$: Reversed about the vertical axis

(4) $f(-3t-2)$: Composite transformation

Add grid lines, x-axis and y-axis labels, and titles for each subplot.

3. Analysis Questions: (2 marks)

- (1) For the signal $f(-3t-2)$, what is the order of transformations? Would the result be the same if the order is changed?
- (2) What is the relationship between the signals $f(3t)$ and $f(t/3)$?

Experiment 1.3: Time-Domain Analysis of Continuous-Time LTI Systems (10 marks)

Objective: To master the basic operations on continuous-time signals, including time shifting, scaling, and reversal, and visualize their effects on signal waveforms using MATLAB.

Given System and Excitation:

The differential equation describing a continuous-time LTI system is given by: $y''(t) + 3y'(t) + 2y(t) = x'(t) + 2x(t)$

The input signal (excitation) to the system is: $x(t) = e^{-2t} u(t)$

Tasks:

- 1. System Modeling:** Based on the differential equation, determine the coefficient vectors for the system transfer function. (2 marks)
- 2. Impulse Response:** Use MATLAB's `impz` function to plot the time-domain waveform of the system's impulse response $h(t)$, suggested time range from $t=0$ to $t=5$ seconds. (2 marks)
- 3. Zero-State Response:** Use MATLAB's `lsim` function to plot the time-domain waveform of the system's zero-state response $y_{zs}(t)$ under the given excitation $x(t)$, suggested time range from $t=0$ to $t=5$ seconds. (2 marks)
- 4. Plot Requirements:** (2 marks)

Plot the impulse response and zero-state response in two separate subplots using the subplot command;

Add grid lines to each plot;

Label the x-axis (time) and y-axis (amplitude);

Add appropriate titles to each plot;

5. Analysis Questions: (2 marks)

- (1) What is the physical meaning of the impulse response $h(t)$? How is it related to the system's transfer function $H(s)$?
- (2) What is the relationship between the zero-state response $y_{zs}(t)$, the impulse response $h(t)$, and the input signal $x(t)$? (Hint: convolution)

Experiment 1.4: Numerical Convolution of Continuous-Time Signals (10 marks)

Objective: Master the method of computing convolution integrals for continuous-time signals using numerical approximation in MATLAB. Understand how the sampling interval affects the accuracy of numerical convolution results.

Given Signals:

Consider two continuous-time signals:

$$1. f_1(t) = \begin{cases} 1.5, & -1 \leq t \leq 1 \\ 0, & \text{Otherwise} \end{cases}$$

$$2. f_2(t) = \begin{cases} 2, & -1 \leq t \leq 1 \\ 0, & \text{Otherwise} \end{cases}$$

Tasks:

1. Analytical Understanding (Optional):

Sketch the convolution $y(t)=f_1(t)*f_2(t)$ analytically and predict its shape.

2. Numerical Convolution with MATLAB: (3 marks)

Using MATLAB, set the sampling interval $p=0.05$ seconds. Discretize both signals and compute their numerical convolution using the **conv** function.

Plot the resulting waveform $y(t)$.

Hint: To approximate continuous convolution, remember to multiply the output of the conv function by the sampling interval p . Also, create a new

time vector for the output that spans from $\min(t_1)+\min(t_2)$ to $\max(t_1)+\max(t_2)$ with step p .

3. Effect of Sampling Interval: (3 marks)

Repeat the numerical convolution for different sampling intervals: $p=0.2, 0.1, 0.05, 0.02$ seconds. Plot all results on the same figure for comparison. Use different line styles or colors to distinguish them.

4. Error Analysis: (2 marks)

Compute the convolution using a very small sampling interval ($p=0.001$) as a reference "exact" solution. For each p value, compute the mean squared error (MSE) and maximum absolute error relative to the reference solution.

5. Discussion Questions: (2 marks)

How does the numerical convolution result change as p decreases?

What is the relationship between p and the accuracy of the numerical approximation?

What are the potential problems if p is too large?

Experiment 2 (20 marks)**Experiment 2.1** Fourier Series Analysis of Periodic Rectangular Pulse Signals (10 marks)

Objective: Understand the spectral characteristics of periodic rectangular pulse signals through Fourier series analysis using MATLAB. Investigate how changes in pulse width affect the frequency spectrum.

Given Signal:

Consider a periodic rectangular pulse signal with amplitude 1, pulse width τ , and period T . The signal is defined as:

$$x(t) = \sum_{n=-\infty}^{\infty} \text{rect}\left(\frac{t - nT}{\tau}\right)$$

where

$$\text{rect}(t) = \begin{cases} 1 & |t| < 0.5 \\ 0 & |t| > 0.5 \end{cases}$$

Tasks:**1. Fourier Series Coefficients Calculation:**

The Fourier series coefficients (exponential form) for the periodic rectangular pulse are given by:

$$c_k = \frac{\tau}{T} \cdot \text{sinc}\left(\frac{k\tau}{T}\right)$$

Where $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$ and $k = 0, \pm 1, \pm 2, \dots$

Special note: For $k=0$, $\text{sinc}(0)=1$, so $c_0 = \frac{\tau}{T}$.

2. MATLAB Implementation:

Write a MATLAB script to compute and plot the magnitude spectrum $|c_k|$ for the following three cases:

Case 1: $\tau=4, T=8$

Case 2: $\tau=2, T=8$

Case 3: $\tau=1, T=8$

Use k values from -20 to 20 for plotting.

3. Plot Requirements:

- (1) Create a 2×2 subplot figure
- (2) Subplot 1: Plot the time-domain waveform for one period of each case (use different colors)
- (3) Subplots 2-4: Plot the magnitude spectra for the three cases
- (4) Label all axes appropriately
- (5) Add grid lines and legends where necessary

4. Analysis Questions:

- (1) How does the pulse width τ affect the width of the main lobe in the frequency domain?
- (2) How does the pulse width τ affect the spacing between spectral lines?
- (3) What happens to the amplitude of the DC component ($k=0$) as τ changes?
- (4) Explain the relationship between time-domain pulse width and frequency-domain bandwidth.

Experiment 2.2: Frequency-Domain Analysis of Continuous-Time LTI Systems (10 marks)

Objective:

Master the frequency-domain method for analyzing continuous-time LTI systems. Use MATLAB to compute the steady-state response of a system with a given frequency response, and observe how the system modifies the input signal.

Given System:

The frequency response of a continuous-time LTI system is:

$$H(j\omega) = \frac{1}{(j\omega)^2 + 2j\omega + 5}$$

Given Input:

The input signal (excitation) is:

$$x(t) = 3 \cos(2t) + 2 \cos(15t)$$

Tasks:

1. Theoretical Analysis (Optional):

Determine the magnitude $|H(j\omega)|$ and phase $\angle H(j\omega)$ at $\omega=2$ rad/s and $\omega=15$ rad/s. Compute the output signal $y(t)$ analytically.

2. MATLAB Implementation:

- (1) Define the frequency response as a function of ω using vectorized operations.
- (2) Evaluate the magnitude and phase of $H(j\omega)$ at the two input frequencies.
- (3) Construct the output signal $y(t)$ as the sum of the scaled and phase-shifted sinusoids.
- (4) Plot the input signal $x(t)$ and the output signal $y(t)$ on the same figure (use two subplots or overlay with different colors).
- (5) Time range: $t=0$ to 10 seconds, with a fine time step (e.g., 0.01 s).

3. Analysis Questions:

- (1) Explain how the system affects the amplitude and phase of each frequency component.
- (2) What type of filter does this system approximate? (low-pass, high-pass, band-pass, band-stop)
- (3) Why does the output waveform look different from the input?

Experiment 3**Experiment 3.1: Frequency-Domain Analysis of an RLC Second-Order Low-Pass Filter (10 marks)****Objective:**

Model a second-order RLC low-pass filter, plot its frequency response (magnitude and phase), and compute the zero-state response to a multi-tone sinusoidal input using MATLAB.

Given Circuit:

The figure shows an RLC series circuit. The input voltage is $x(t)=f(t)$, and the output voltage $y(t)$ is taken across the capacitor. The component values are: Inductor $L=0.6\text{H}$, Capacitor: $C=0.2\text{F}$, Resistor: $R=2\Omega$.

System Function:

The transfer function of this filter is:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Tasks:**1. Frequency Response Plotting:**

Use MATLAB's `freqs()` function to compute and plot the magnitude response $|H(j\omega)|$ and phase response $\angle H(j\omega)$ over a suitable frequency range (e.g., $\omega=0$ to 20 rad/s).

Use `abs()` for magnitude and `angle()` for phase (convert to degrees).

2. Zero-State Response:

The input signal is $x(t) = \sin(t) + \sin(20t)$.

Use the `lsim()` function to compute and plot the zero-state response $y_{zs}(t)$ over the time interval $t=0$ to 10 seconds.

3. Plot Requirements:

- (1) Create one figure with two subplots: magnitude response (top) and phase response (bottom).
- (2) Create a second figure showing the input signal $x(t)$ and the output signal $y_{zs}(t)$ on the same axes (use different colors).
- (3) Label all axes and add grids.

4. Analysis Questions:

- (1) What type of filter is this? Estimate its cutoff frequency from the magnitude plot.
- (2) Compare the input and output waveforms. Which frequency component is attenuated more? Why?
- (3) How would the response change if the resistor value R were increased?

Experiment 3.2: Amplitude Modulation, Coherent Demodulation, and Low-Pass Filtering (10 marks)

Objective:

Simulate an amplitude modulation (AM) and coherent demodulation system using MATLAB. Observe the time-domain waveforms and frequency spectra of the signals at each stage, and verify the function of the low-pass filter in recovering the baseband signal.

System Description:

1. Baseband signal (modulating signal):

$$g(t)=4\cos(8t) + 3\cos(16t)$$

2. Carrier signal:

$$c(t)=\cos(120t)$$

3. Modulated signal (AM-DSB-SC):

$$f(t)=g(t)\cdot c(t)$$

4. Coherent demodulation:

$$g_0(t)=f(t)\cdot c(t)$$

5. Low-pass filter:

Ideal low-pass filter with frequency response:

$$H(j\omega) = \begin{cases} 1, & |\omega| < 25 \\ 0, & \text{Otherwise} \end{cases}$$

The filtered output is $g_1(t)$.

Tasks:**1. Time-domain waveform plotting:**

Generate the following signals over the time interval $t=0$ to 22 seconds (use a sampling frequency of at least 1000 Hz):

- (1) Baseband signal $g(t)$
- (2) Modulated signal $f(t)$
- (3) Demodulated signal (before filtering) $g_0(t)$
- (4) Recovered signal (after filtering) $g_1(t)$

Plot all four waveforms in one figure (use subplot or overlay with clear legends). Label axes and add titles.

Hint: You can implement this ideal low-pass filter in the frequency domain.

First, compute the FFT of $g_0(t)$ and use `fftshift` to center the zero frequency.

Then set the Fourier coefficients corresponding to frequencies $|\omega| \geq 25$ to zero, use `ifftshift`, and finally use the `ifft` function to transform it back to the time domain.

2. Frequency spectrum plotting:

Compute the magnitude spectra (using FFT) of the four signals above.

Plot all four spectra in another figure (use subplot).

Normalize the frequency axis to rad/s and clearly indicate the frequency components.

3. Analysis questions:

Explain why the modulated signal $f(t)$ contains frequency components at 120 ± 8 and 120 ± 16 rad/s.

Why does the demodulated signal $g_0(t)$ contain both baseband components and high-frequency components around 240 rad/s?

How does the low-pass filter recover $g(t)$? What would happen if the filter cutoff frequency were set below 16 rad/s?

Compare the original baseband signal $g(t)$ and the recovered signal $g_1(t)$. Are they identical? Explain any differences.

Experiment 4

Experiment 4.1: Geometric Evaluation of Frequency Response from Pole-Zero Plot (10 marks)

Objective:

Given the pole-zero locations of a second-order system, use the geometric method (vector lengths from poles/zeros to points on the imaginary axis) to compute and plot the magnitude response over 0–1 kHz. Identify the type of filter and explain its function.

System Description:

A second-order continuous-time LTI system has the following poles and zeros:

Poles: $p_1 = -80, p_2 = -120$

Zeros: $z_1 = z_2 = 0$ (double zero at the origin)

Assume the system gain $K=1$.

Tasks:

1. Pole-Zero Plot:

Plot the pole-zero diagram of the system in the s-plane.

2. Geometric Evaluation of Frequency Response:

For frequencies $f=0$ to 1000 Hz (step size ≤ 1 Hz), compute the magnitude response $|H(j\omega)|$ using the geometric method:

$$|H(j\omega)| = \frac{|j\omega - z_1| \cdot |j\omega - z_2|}{|j\omega - p_1| \cdot |j\omega - p_2|}$$

Where $\omega=2\pi f$.

3. Plotting:

Plot the magnitude response (linear or dB scale) versus frequency (Hz).

Add grid lines, label axes, and provide a title.

4. Analysis Questions:

- (1) What type of filter is this? (low-pass, high-pass, band-pass, band-stop, etc.)
- (2) Explain the shape of the magnitude response using the geometric interpretation of poles and zeros.
- (3) How would the response change if the zeros were moved away from the origin (e.g., to real-axis positions)?
- (4) Verify your result using MATLAB's `freqs()` function and compare with the geometric method.

Experiment 4.2 (10 marks)

Given the system function:

$$H(s) = \frac{s^2 + 4s + 3}{s^3 + s^2 + 7s + 2}$$

use the MATLAB command pzmap to plot its pole-zero map and determine whether the system is stable based on the pole locations.